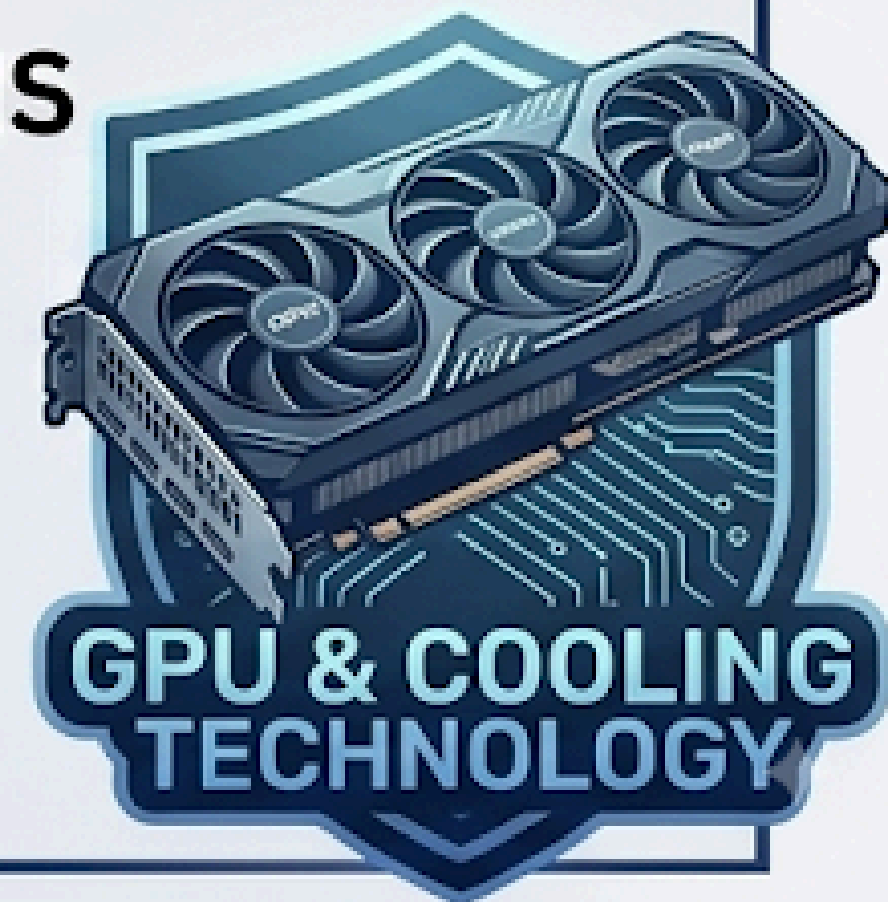


NAME:-HET ACHARYA

Course: Diploma in Cyber Security

DAY 4 – GPUs AND COOLING SYSTEMS

Topic: GPUs and Cooling Systems



WHAT IS A GPU?

GPU – Graphics Processing Unit

A GPU is a processor that mainly handles graphics and visual calculations.

It helps the computer create the images, videos, animations and game graphics that we see on the screen.

Simple example

When we play a graphics-heavy game, the GPU helps generate:

- Characters
- Buildings
- Shadows
- Lighting
- Textures
- Game environments

Easy memory trick:

CPU → General processing

GPU → Graphics processing

INTEGRATED GPU VS DEDICATED GPU

Integrated GPU

Built into CPU/package

Usually uses system RAM

Lower power use

Good for basic graphics

Usually cheaper

Dedicated GPU

Separate graphics card

Usually has dedicated VRAM

Usually higher power use

Better for demanding graphics

Usually more expensive

1. Integrated GPU

An integrated GPU is built into the CPU or processor package and normally shares system memory with the CPU.

It is suitable for everyday tasks such as:

- Browsing
- Videos
- Office work
- Basic gaming

Integrated = Built in

2. Dedicated GPU

A dedicated GPU is a separate graphics processor, usually installed as an expansion card in a desktop PC.

It has its own graphics memory called VRAM and is designed for more demanding graphics workloads.

It is useful for:

- High-end gaming
- 3D work
- Video editing
- AI/graphics workloads

Dedicated = Separate

HOW TO IDENTIFY A GPU

GPU Identification

For the practical task, check the computer's hardware information and note:

1. Brand

Examples:

NVIDIA, AMD, Intel

2. Model

Examples:

GeForce RTX 4060

Radeon RX 7600

Intel Arc A750

3. Type

Write whether it is:

Integrated or Dedicated

Example

Brand: NVIDIA

Model: GeForce RTX 4060

Type: Dedicated GPU

Important:

Do not write this as your own GPU unless your computer actually has this model.

For your assignment, replace it with the GPU that is actually installed on your PC/laptop.

Where can I check it?

On Windows:

Task Manager → Performance → GPU

You can also check:

Device Manager → Display adapters

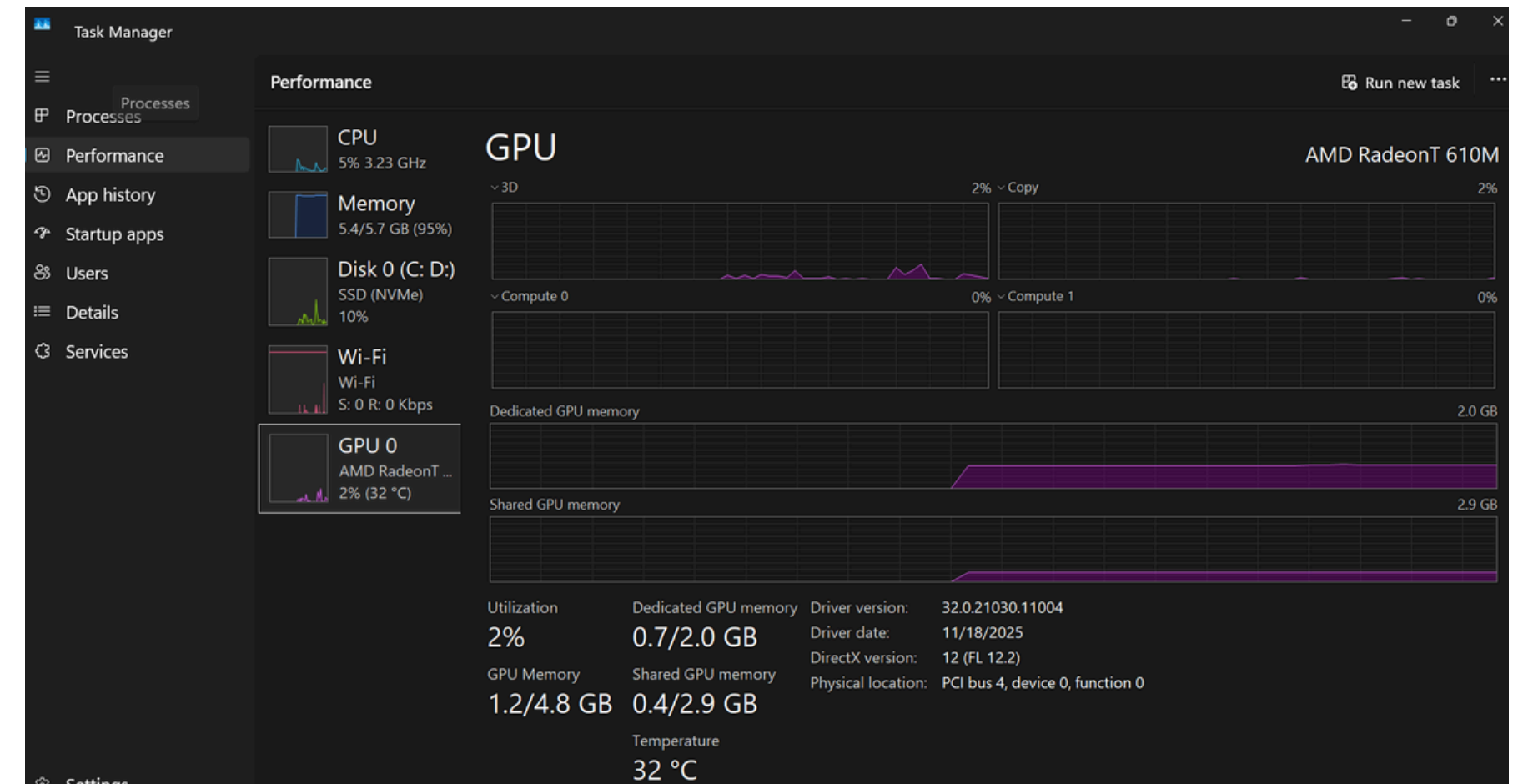


Figure 1: GPU information of my computer

WHY DOES A GPU NEED COOLING?

GPU + Heat

When a GPU performs heavy work, it uses electrical power and produces heat. Gaming and other demanding workloads can make the GPU work harder, so proper cooling is important.

Why is cooling needed?

Cooling helps:

- Remove excess heat
- Keep temperatures under control
- Maintain stable performance
- Protect hardware from excessive heat

What happens if a GPU gets too hot?

The GPU can reduce its operating performance to control temperature. This is commonly called thermal throttling.

Easy memory:

More GPU work → More heat → Cooling removes heat

MAIN COOLING COMPONENTS

1. Fan

A fan moves air through or around the cooling system.

Its job is to help carry heat away.

Easy memory: Fan = Moves air

2. Heatsink

A heatsink is a metal component that absorbs heat from the GPU or another chip and spreads it over a larger surface area.

The fan then helps move air across it.

Easy memory: Heatsink = Spreads heat

3. Thermal Paste

Thermal paste fills tiny gaps between the chip and heatsink so heat can transfer more efficiently.

Easy memory: Thermal paste = Helps heat transfer

4. Radiator

A radiator is used in many liquid cooling systems. It transfers heat from the liquid to the surrounding air.

Easy memory: Radiator = Releases heat

5. Pump

In a liquid cooling system, the pump circulates the cooling liquid through the system.

Easy memory: Pump = Moves liquid

Simple cooling flow:

GPU → Heatsink/Cold plate → Heat carried away → Fan/Radiator → Air

AIR COOLING VS LIQUID COOLING

Feature	Air Cooling	Liquid Cooling
Main method	Heatsink + fan	Coolant + pump + radiator
Cost	Usually lower	Usually higher
Installation	Simpler	More complex
Maintenance	Generally easier	More components to maintain
Failure points	Fewer	More, such as pump
Noise	Depends on fans	Can be quiet, but also uses fans/pump
Best for	Most normal PCs	Some high-performance setups

- Air Cooling

Air cooling uses a heatsink and fan to move heat away from the component.

Good points:

• Simple

• Usually cheaper

• Easy to maintain

• Fewer components
- Liquid Cooling

Liquid cooling uses coolant, a pump and a radiator to move heat away from the component.

Good points:

• Can handle high heat loads

• Can be useful in high-performance systems

• Can offer more flexibility in some PC designs

Easy memory trick:

Air = Simple

Liquid = More complex

WHICH COOLING SHOULD I CHOOSE?

Scenario: Gaming + IPL Streaming PC

Imagine I am building a PC for:

- Watching IPL matches
- Streaming
- Playing graphics-heavy games
- Long gaming sessions

My choice: AIR COOLING

I would choose air cooling for a normal gaming PC because it is simple, reliable and generally easier to maintain.

A good GPU cooler and proper case airflow can keep a gaming system at suitable temperatures without adding unnecessary complexity.

Liquid cooling can also be useful in some high-performance systems, but I would choose it when there is a specific reason for it rather than using it just because it looks advanced.

My 3-line answer:

“For my gaming and streaming PC, I would choose air cooling. It is simpler, generally cheaper and easier to maintain. If the system had unusually high cooling requirements, I would consider liquid cooling instead.”

PRACTICAL: IDENTIFYING THE COOLING SYSTEM

What should I look for?

When looking inside a desktop PC, look for:

GPU

A dedicated GPU usually looks like a large expansion card installed into a PCIe slot.

GPU fans

Many dedicated graphics cards have one or more fans attached to their cooling system.

Heatsink

Look for the metal fin structure that helps spread heat.

Case fans

These move air through the PC case.

CPU cooler

Do not confuse the CPU cooler with the GPU cooler. Both may have fans and heatsinks, but they cool different processors.

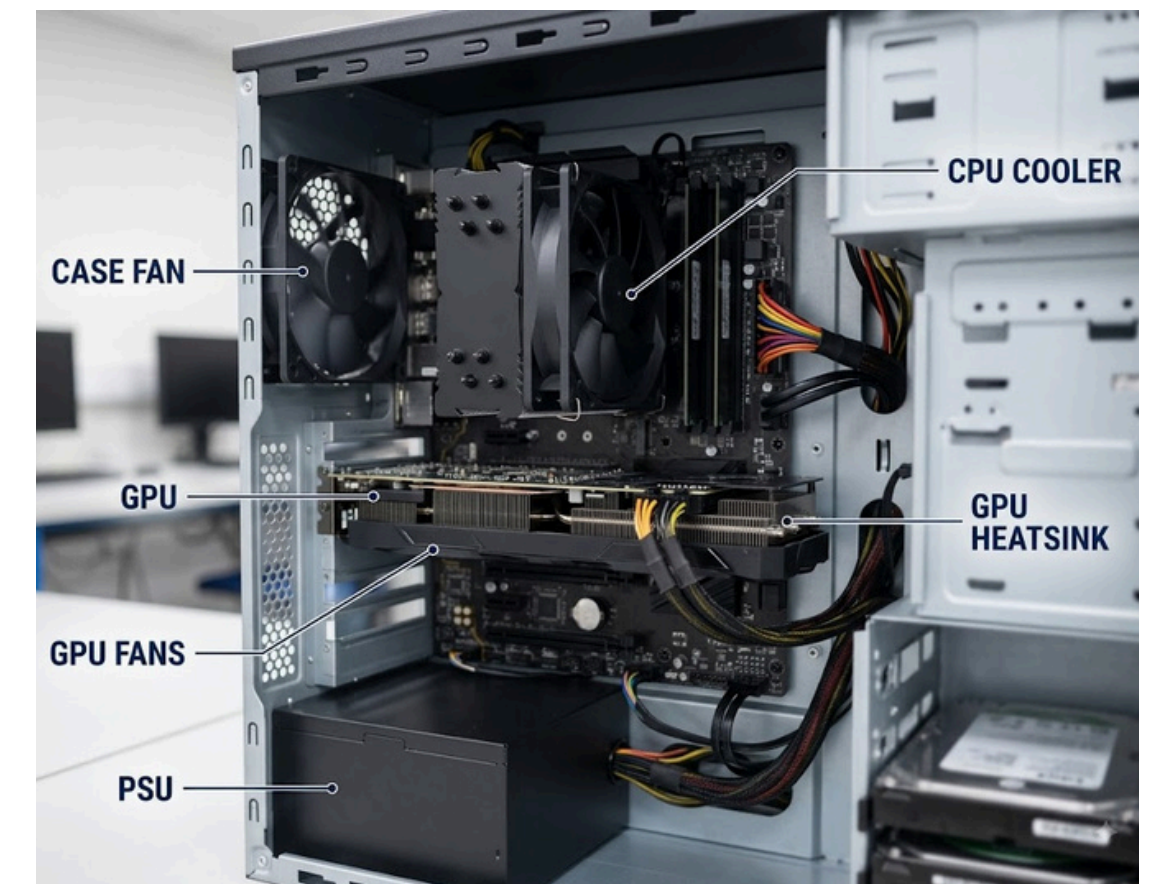


Figure 2: GPU and cooling components inside a PC

CONCLUSION

In this practical, I learned about GPUs and their role in computer systems. A GPU is responsible for handling graphics and visual processing, which is very important for gaming, video editing and other graphics-heavy tasks.

I also learned the difference between integrated and dedicated GPUs, where integrated GPUs are built into the processor platform and dedicated GPUs are separate components with their own VRAM.

Another important concept I learned is cooling systems. GPUs generate heat during operation, and cooling systems like fans, heatsinks and thermal paste help maintain safe operating temperatures. In some advanced systems, liquid cooling is also used.

Final understanding:

A GPU improves graphics performance, but a proper cooling system is equally important to maintain stability and prevent overheating.